

3.5 Carbonyl Compounds

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.5 Carbonyl Compounds
Difficulty	Hard

Time allowed: 40

Score: /30

Percentage: /100

Question 1a

Aqueous sodium tetrahydridoborate, NaBH_4 , is a common reducing agent.

Identify **two** isomers with the formula $\text{C}_3\text{H}_6\text{O}$ that cannot be reduced by aqueous NaBH_4 .

[2 marks]

Question 1b

Identify the **two** isomers with the formula $\text{C}_3\text{H}_6\text{O}$ that can be reduced by aqueous NaBH_4 .

[2 marks]

Question 1c

When NaBH_4 is used as a reducing agent followed by the addition of acid, the reduction products of ketones can exhibit optical isomerism, while the reduction products of aldehydes cannot.

i)

Classify the reduction products of aldehydes and ketones.

[2]

ii)

Explain why the reduction products of ketones can exhibit optical isomerism, while the reduction products of aldehydes cannot.

[2]

[4 marks]

Question 1d

Explain why the reduction product of a carbonyl compound, by NaBH_4 and acid, cannot be a tertiary alcohol.

[2 marks]

Question 2a

This question is about the reactions of an organic compound **R**.

Analysis of 15.0 g of an organic compound, **R**, showed it to contain 69.8% carbon, 2.79 g of oxygen and the remaining mass was hydrogen.

- i)
Calculate the empirical formula of the organic compound **R**.
- ii)
Compound **R** has a molecular mass of 86.0 g mol^{-1} , deduce the molecular formula of compound **R**.

[4 marks]

Question 2b

Compound **R** is a straight chain organic compound. One isomer of compound **R** produces optical isomers of compound **S** when reacted with hydrogen cyanide followed by dilute acid.

Compounds **R** and **S** were tested with acidified potassium dichromate(VI) and Tollens reagent. The results are shown in **Table 2.1**.

Table 2.1

	R	S
Acidified potassium dichromate(VI)	Green solution	Green solution
Tollens reagent	Silver mirror formed	No visible change

Using this information and your answer from part **(a)**, identify compounds **R** and **S**.
Justify your answer.

[4 marks]**Question 2c**

Draw 3D representations of the two optical isomers of compound **S**.

[1 mark]**Question 2d**

There are some isomers of compound **S** that do not display optical isomerism.

Draw the skeletal formula of these isomers.

[1 mark]

Question 3a

LiAlH_4 is a reducing agent.

LiAlH_4 cannot be used in aqueous solution because it reacts with water to produce LiOH(aq) , $\text{H}_2(\text{g})$ and a white precipitate which is soluble in excess sodium hydroxide.

Identify the white precipitate.

[1 mark]

Question 3b

Two students try to prepare 2-hydroxybutanoic acid in the laboratory as shown in Fig. 3.1

Both students oxidise butane-1,2-diol to form **P** in reaction 1.

One student then reduces **P** using LiAlH_4 . **Q** is formed.

The other student reduces **P** using NaBH_4 . **R** is formed.

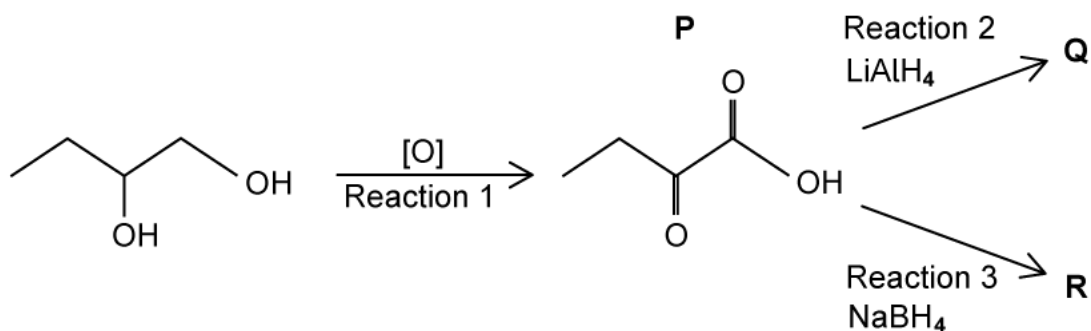


Fig. 3.1

i)

State the reagents and conditions required for reaction 1.

Only one of the students successfully prepares 2-hydroxybutanoic acid.

[2]

ii)

Identify which of **Q** or **R** is 2-hydroxybutanoic acid and explain the difference between reactions 2 and 3.

[2]

[4 marks]

Question 3c

A third student prepares 2-hydroxybutanoic acid using propanal as the starting material as shown in Fig. 3.2. In step 1 the student reacts propanal with a mixture of NaCN and HCN.

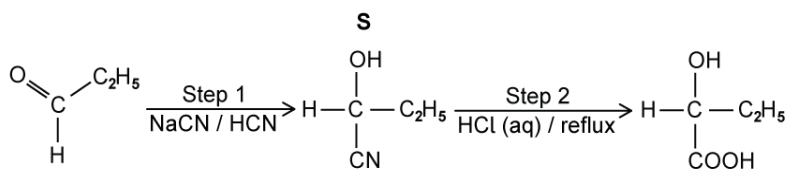


Fig. 3.2

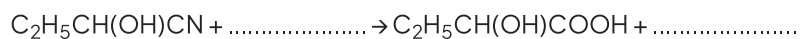
Draw the mechanism for the reaction of propanal with the mixture of NaCN and HCN to form **S**.

- Identify the ion that reacts with propanal.
- Draw the structure of the intermediate of the reaction.
- Include all charges, partial charges, lone pairs and curly arrows.

[4 marks]

Question 3d

Complete the equation for the reaction in step 2, when **S** is heated under reflux with HCl(aq).



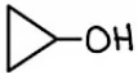
[1 mark]

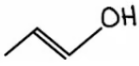
Model Answers: Hard

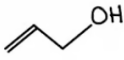
1a

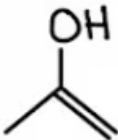
a) **Two** isomers with the formula C_3H_6O that cannot be reduced by aqueous $NaBH_4$ are:

Any **two** of the following

Cyclopropanol **OR** 
; [1 mark]

Prop-1-en-1-ol **OR** 
; [1 mark]

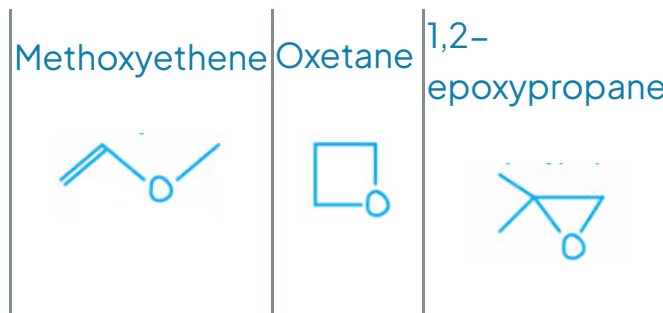
Prop-2-en-1-ol **OR** 
; [1 mark]

Prop-2-en-2-ol **OR** 
; [1 mark]

[Total: 2 marks]

- You would also score the mark for the correct displayed or structural formulae
- The fact that the formula has an O suggests carbonyls, alcohols, epoxides and ethers
- **Remember:** Although the following chemicals are possible options, epoxides and ethers are not specified on the syllabus

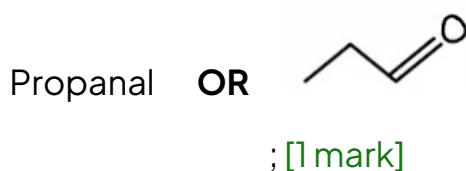




- Carbonyls can be reduced by aqueous NaBH_4
- Carbonyls form the corresponding alcohol when reduced by aqueous NaBH_4
- Therefore, you should consider alcohols with the formula $\text{C}_3\text{H}_6\text{O}$
- The fact that the carbon and hydrogen portion of the formula matches the general formula for an alkene, C_nH_{2n} , suggests
 - Either an alkene-based structure
 - Or a cyclic / ring-based structure

1b

b) The **two** isomers with the formula $\text{C}_3\text{H}_6\text{O}$ that can be reduced by aqueous NaBH_4 are:



[Total: 2 marks]

- To be reduced by NaBH_4 , the isomers **must** be an aldehyde or a ketone
- This means that there is only one option for the aldehyde and one for the ketone that fit the $\text{C}_3\text{H}_6\text{O}$ formula

1c

c)

i) The reduction products are:

- Aldehydes become primary alcohols; [1 mark]
- Ketones become secondary alcohols; [1 mark]

ii) The reduction products of ketones can exhibit optical isomerism, while the reduction products of aldehydes cannot because:

- Primary alcohols have two hydrogens attached to the alcoholic carbon
AND
(Therefore,) primary alcohols do not have a chiral carbon; [1 mark]
- Secondary alcohols can have three different functional groups attached to the alcoholic carbon
AND
(Therefore,) secondary alcohols can have a chiral carbon; [1 mark]

[Total: 4 marks]

- Remember:** An optical isomer requires a chiral carbon – a carbon with four different groups attached
- Aldehydes cannot produce optical isomers when reduced by NaBH_4 as the reduction product will have two hydrogens
- Ketones can form optical isomers **if** the reduction product has four different groups attached to the alcoholic carbon
- Propanone does **not** form optical isomers as the alcoholic carbon will have two methyl groups attached

1d

d) The reduction product of a carbonyl compound cannot be a tertiary alcohol because:

- Tertiary alcohols have the general formula R_3COH
AND
Carbonyl compounds have the general formula $\text{R}_2\text{C}=\text{O}$; [1 mark]
- When a ketone / R_2CO is reduced (by NaBH_4) it becomes R_2CHOH

AND

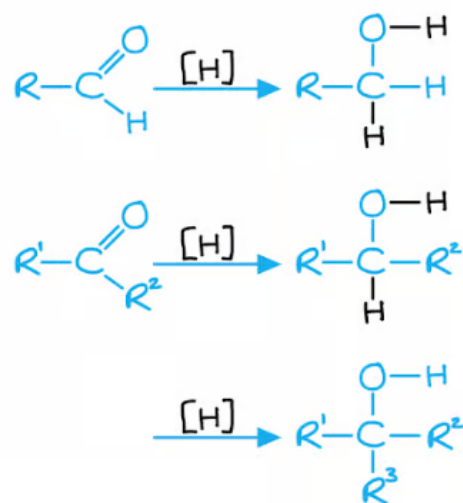
When an aldehyde / RCHO is reduced (by NaBH_4) it becomes RCH_2OH

AND

The presence of the C-H bond formed means it cannot be a tertiary alcohol; [1 mark]

[Total: 2 marks]

- As previously mentioned, aldehydes become primary alcohols and ketones become secondary alcohols



- If moving forwards to form a tertiary alcohol was reduction, then moving backwards would be oxidation and you should know that tertiary alcohols do not undergo oxidation

2a

a)

i) To calculate the empirical formula of the organic compound **R**:

- Calculate the mass of carbon and hydrogen in the 15.0 g sample

$$\text{Carbon: } 15.0 \times \frac{69.8}{100} = 10.47 \text{ g}$$

AND

$$\text{Hydrogen: } 15.0 - 10.47 - 2.79 = 1.74 \text{ g; [1 mark]}$$

- Divide the mass by the relevant A_r ; [1 mark]

$$C = \frac{10.47}{12.0} \quad H = \frac{1.74}{1.0} \quad O = \frac{2.79}{16.0}$$

$$= 0.8725 \quad = 1.74 \quad = 0.174$$

- Divide each number by the smallest number (0.174) calculated to get the ratio

$$C = 5 : H = 10 : O = 1$$

AND

Therefore, the empirical formula is $C_5H_{10}O$; [1 mark]

ii) To deduce the molecular formula of compound **R**:

- Calculate the M_r of the empirical formula

$$(12.0 \times 5) + (1.0 \times 10) + (16.0 \times 1) = 86.0$$

AND

Therefore the molecular formula is the empirical formula / $C_5H_{10}O$; [1 mark]

[Total: 4 marks]

- The key to this question is how you get the individual elements into the same units for use in the empirical formula calculation
- You can do this by converting the percentage of carbon to a mass as shown above
- An alternative method involves calculating the percentage of the sample that is oxygen:
 - $(\frac{2.79}{15.0} \times 100 = 18.6\%)$, deducing the percentage of hydrogen
 $(100 - 69.8 - 18.6 = 11.6\%)$
- You can then complete the empirical formula calculation using the percentage values

2b

b) To identify compounds **R** and **S**:

- Compounds **R** and **S** react with acidified potassium dichromate(VI), therefore they could be an aldehyde / a primary alcohol / a secondary alcohol; [1 mark]
- Compound **R** reacts with Tollens reagent

AND

Compound **R** has the formula $C_5H_{10}O$, therefore it is pentanal; [1 mark]

- Compound **S** reacts with acidified potassium dichromate(VI) but does not react with Tollens reagent, therefore it must be a primary alcohol or a secondary alcohol; [1 mark]

- Compound **S** is formed by the reaction of compound **R** / pentanal with HCN

AND

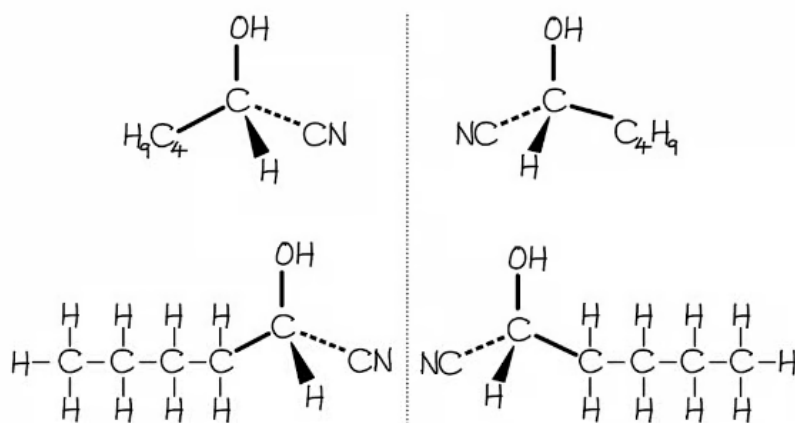
Compound **S** is 2-hydroxyhexanenitrile; [1 mark]

[Total: 4 marks]

- **Careful:** With a question like this, it is very easy to forget that you have already calculated the molecular formula because you become so engrossed in interpreting the results of the chemical tests
- **Remember:** Aldehydes can give positive results with acidified potassium dichromate(VI) solution
- As compound **R** forms compound **S**, you have to deduce the identity of one compound to be able to get the other
- The positive result of compound **R** with Tollens reagent tells you it is an aldehyde and with a formula of $C_5H_{10}O$ and the knowledge that **R** is a straight chain, it **must** be pentanal
- After this, you have to work/think your way through the nucleophilic addition of cyanide to form the nitrile compound that is compound **S**
- **Careful:** The name of the nitrile compound **must** have the “e” in the alkane chain
- E.g. 2-hydroxyhexanenitrile does get the mark
- E.g. 2-hydroxyhexan_nitrile does not get the mark

2c

c) The 3D representations of the two optical isomers of compound **S** are:



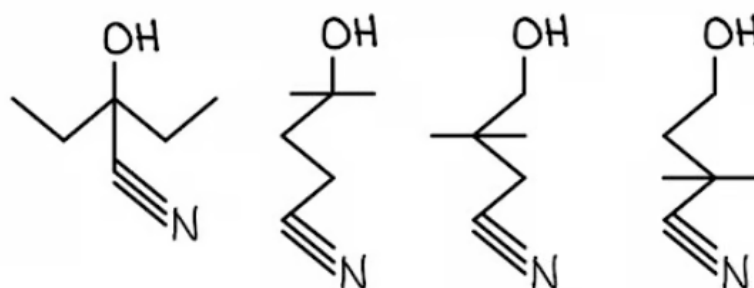
- Correct 3D representations of both isomers; [1 mark]

[Total: 1 mark]

- From part (c) you have already identified that compound **S** is 2-hydroxyhexanenitrile
- When optical isomerism is mentioned your first thought should be to identify the chiral carbon in the molecule
- By drawing out the displayed structure of compound **S** can help you see the chiral carbon with four different groups attached
- You should show at least one wedge bond and one dash bond in each structure
- You should have no bonds at 180° to each other

2d

d) The skeletal formulae of the compound **S** isomers that do not display optical isomerism are:

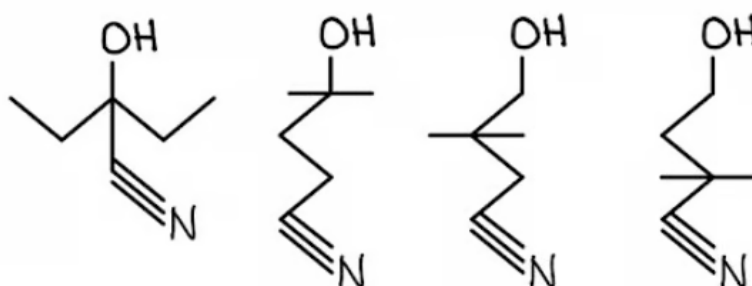


[Total: 1 mark]

- From part (c) you have already identified that compound **S** is 2-hydroxyhexanenitrile
- When optical isomerism is mentioned your first thought should be to identify the chiral carbon in the molecule
- By drawing out the displayed structure of compound **S** can help you see the chiral carbon with four different groups attached
- You should show at least one wedge bond and one dash bond in each structure
- You should have no bonds at 180° to each other

2d

d) The skeletal formulae of the compound **S** isomers that do not display optical isomerism are:



- Any two correct structures; [1 mark]

[Total: 1 mark]

- The molecular formula of compound **S** is $C_6H_{11}NO$ and this includes an $-OH$ functional group and a $-CN$ functional group
- **Remember:** Optical isomers have four **different** atoms or groups of atoms on the asymmetrical carbon
- **Careful:** You are asked for the **skeletal** structures that do **not** show optical isomerism
- This means that you are checking that none of your proposed structures contain carbons with four different atoms or groups of atoms
- Therefore, all of the carbons in your proposed structures should have **two of the same** atoms, e.g. hydrogens, or groups of atoms, e.g. methyl / ethyl groups, on the carbon atoms

3a

a) The white precipitate is:

- Aluminium hydroxide / $\text{Al}(\text{OH})_3$; [1 mark]

[Total: 1 mark]

- You are told that lithium hydroxide is formed
- It is logically to deduce that the other metals in the LiAlH_4 will also react with water to form a hydroxide, $\text{Al}(\text{OH})_3$
- From GCSE you should know that aluminium ions form a white precipitate in sodium hydroxide, and that in excess NaOH the precipitate should redissolve

3b

b)

i) The reagents and conditions required for reaction 1 are:

- Potassium dichromate (VI); [1 mark]
- Acid(ified)

AND

Heat under reflux; [1 mark]

ii) Identify which of **Q** or **R** is 2-hydroxybutanoic acid and explain the difference between reactions 2 and 3:

- R** is 2-hydroxybutanoic acid
- AND**
- (Because) Only the $\text{C}=\text{O}$ / ketone is reduced; [1 mark]
- NaBH_4 cannot reduce the COOH / carboxylic acid; [1 mark]

[Total: 4 marks]

- In reaction 1, a diol is being oxidised (shown by $[\text{O}]$) so the reagent must be an oxidising agent
- You would also score the mark for potassium manganate (VII)
- P** contains a carbonyl group and a carboxylic acid group
- When the compound is reduced using LiAlH_4 , both carbonyl and carboxylic acid

group would gain hydrogen

- When the compound is reduced using NaBH_4 only the carbonyl group will gain hydrogen
- LiAlH_4 is stronger reducing agent than NaBH_4
- You are expected to know that an aldehyde / ketone can be reduced using both of these substances, but only LiAlH_4 is the reducing agent for a carboxylic acid

3c

c) The mechanism for the reaction of propanal with the mixture of NaCN and HCN to form **S** is:



- Presence of $:\text{CN}^-$; [1 mark]
- Curly arrow from $:\text{CN}^-$ lone pair to carbonyl carbon; [1 mark]
- Correct dipole **AND** curly arrow from double bond to oxygen; [1 mark]
- Correct intermediate drawn; [1 mark]

[Total: 4 marks]

- If you show the bonds within the cyanide ion, it must clearly be a triple bond or you will not score the mark
- NaCN acts as catalyst in the reaction whilst the HCN provide the nucleophile, $:\text{CN}^-$ for the reaction
- You will score the marks if your negative charges are not circled

3d

d) The equation for the reaction in step 2, when **S** is heated under reflux with HCl (aq) is:

